

Determining the Number of Sides of Polygons

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The sum of the **exterior angles** of any polygon
is always 360° .

Exterior Angle of a Regular Polygon

For a regular polygon, the measure of one exterior angle is:

$$\text{Exterior Angle} = \frac{360^\circ}{n}$$

where n is the number of sides.

Finding the Number of Sides

To determine the number of sides when the exterior angle is known:

$$n = \frac{360^\circ}{\text{Exterior Angle}}$$

Interior Angle

The measure of one **interior angle** of a regular polygon is:

$$\text{Interior Angle} = 180^\circ - \text{Exterior Angle}$$

Finding the Number of Sides

If the interior angle is known, first find the exterior angle:

$$\text{Exterior Angle} = 180^\circ - \text{Interior Angle}$$

Then:

$$n = \frac{360^\circ}{180^\circ - \text{Interior Angle}}$$

Examples

Example 1

The exterior angle of a regular polygon measures 45° . Determine the number of sides.

$$n = \frac{360^\circ}{45^\circ} = 8$$

The polygon has 8 sides.

Example 2

The exterior angle of a regular polygon measures 30° . Determine the number of sides.

$$n = \frac{360^\circ}{30^\circ} = 12$$

The polygon has 12 sides.

Example 3

The interior angle of a regular polygon measures 150° . Determine the number of sides.

$$E = 180^\circ - 150^\circ = 30^\circ$$

$$n = \frac{360^\circ}{30^\circ} = 12$$

The polygon has 12 sides.

Practice Exercises

Practice 1

The exterior angle measures 24° . Determine the number of sides.

Practice 2

The exterior angle measures 15° . Determine the number of sides.

Practice 3

The exterior angle measures 12° . Determine the number of sides.

Practice 4

The interior angle measures 120° . Determine the number of sides.

Practice 5

The interior angle measures 135° . Determine the number of sides.

Practice 6

The interior angle measures 160° . Determine the number of sides.

Practice 7

The interior angle measures 165° . Determine the number of sides.

Activities

Activity 1

The exterior angle measures 60° . Determine the number of sides.

Activity 2

The exterior angle measures 40° . Determine the number of sides.

Activity 3

The exterior angle measures 20° . Determine the number of sides.

Activity 4

The exterior angle measures 18° . Determine the number of sides.

Activity 5

The exterior angle measures 10° . Determine the number of sides.

Activity 6

The interior angle measures 108° . Determine the number of sides.

Activity 7

The interior angle measures 140° . Determine the number of sides.

Activity 8

The interior angle measures 144° . Determine the number of sides.

Activity 9

The interior angle measures 156° . Determine the number of sides.

Activity 10

The interior angle measures 170° . Determine the number of sides.

Thank you!